# HUAN NGUYEN

Academic Curriculum Vitae

v.huannd13@vinrobotics.net | Google Scholar | LinkedIn | huannguyenarl.github.io

## Education

2023 **PhD in Engineering Cybernetics**
Norwegian University of Science and Technology (NTNU), Norway.
Research in robotics; supervisor: Prof. Kostas Alexis.

2020 **MS in Computer Science and Engineering**
University of Nevada, Reno, USA.
Thesis: Aerial Robotic Chain: Modeling, Control, Shape and Motion Planning.

2018 **Master's degree in Electrical and Electronics Engineering**
Ho Chi Minh City University of Technology, Vietnam.

2016 **Bachelor's degree in Electrical and Electronics Engineering**
Ho Chi Minh City University of Technology, Vietnam.

## Research Interests

Embodied AI and autonomous systems; model predictive control; uncertainty-aware path and motion planning; multi-aerial-robot systems; aerial manipulation; multimodal SLAM and state estimation; robot learning and vision-language navigation; field robotics; robotics hardware and system integration.

## Research and Industry Experience

**Head of SLAM/NAV Department, VinRobotics | Mar 2025–Present**

- Lead a 10-person department researching and developing robot localization, mapping and navigation.
- Design LiDAR–inertial–leg odometry SLAM for humanoid robots in industrial environments; co-lead vision-language navigation research for high-level reasoning.

**Lead Autonomy Engineer, Hyke | 2024–2025**

- Led guidance, navigation and control development for electric ferries, including system identification, dynamic positioning, control allocation, and path and motion planning.
- Performed sensor calibration for state estimation and autonomous operation.

**PhD Researcher, Autonomous Robots Lab, NTNU | Jan 2021–2023**

- Developed uncertainty-aware, learning-based navigation for aerial robots and validated the methods in diverse field environments.

**Graduate Research Assistant, Autonomous Robots Lab, University of Nevada, Reno | Aug 2018–Dec 2020**

- Developed modelling, control, and shape and motion planning for reconfigurable aerial robotic systems capable of forceful physical interaction.

## Selected Research Contributions

**Uncertainty-aware informative path planning and learned navigation.** Integrated learned collision and information-gain prediction for uncertainty-aware informative path planning, validated in real-world aerial-robot experiments [J1]. Related work includes motion-primitive planning, semantic-enhanced collision prediction, and risk-aware navigation [C2, C5, C8].

**Control and aerial manipulation.** Developed hybrid modelling and control for a reconfigurable aerial robotic chain in free flight and physical interaction modes [J4, C16], alongside collaborative work on shape and motion planning [C15]. First-authored a survey of model predictive control for micro aerial vehicles [C9].

**Resilient perception and field autonomy.** Contributed to complementary multimodal sensor fusion for robust localization in degraded subterranean environments [C14]. Field robotics work spans collision-tolerant aerial platforms, autonomous exploration and infrastructure inspection [C1, C6, C10, C11, C19, C20].

**Multi-robot systems and hardware integration.** Contributed to teamed exploration with legged and aerial robots [C4, C7] and to a self-deployed wireless communications system for subterranean robots [C18]. In Team CERBERUS, co-responsible for the autonomy stack of collision-tolerant aerial robots and developed LiDAR–camera calibration software.

## Awards and Recognition

- **ICAR 2023 Best Paper Award.** Co-author of "Autonomous Exploration and General Visual Inspection of Ship Ballast Water Tanks Using Aerial Robots" [C1].
- **IROS 2022 Best Paper Award on Safety, Security, and Rescue Robotics (finalist).** Co-author of "Risk-Aware Motion Planning for Collision-Tolerant Aerial Robots Subject to Localization Uncertainty" [C5].
- **IEEE ICRA 2022 Outstanding Deployed Systems Paper Award.** Co-author of "Autonomous Teamed Exploration of Subterranean Environments Using Legged and Aerial Robots" [C7].
- **DARPA Subterranean Challenge, 2021**. Member of the winning CERBERUS team (2018–2021).

## Teaching and Mentoring

**Mentor, VinRobotics | Mar 2025–Present**

- Mentor junior engineers and interns on SLAM and navigation projects, from problem formulation to implementation and real-robot testing.

**Teaching Assistant, University of Nevada, Reno | Jan 2019–May 2020**

- Undergraduate courses: Data Structures; Mathematics of Computer Science.

**Teaching interests:** robot perception and state estimation; path and motion planning; robot dynamics and control, including model predictive control; experimental robotics and autonomous systems.

## Publications

Long consortium author lists are abbreviated.

### Journal Articles

**[J1] H. Nguyen**, R. Andersen, E. Boukas and K. Alexis. Uncertainty-aware visually-attentive navigation using deep neural networks. The International Journal of Robotics Research, 43(6), 840–872, 2024.

**[J2]** M. Tranzatto et al. (including **H. Nguyen**). Team CERBERUS Wins the DARPA Subterranean Challenge: Technical Overview and Lessons Learned. Field Robotics, 4, 249–312, 2024.

**[J3]** M. Tranzatto et al. (including **H. Nguyen**). CERBERUS: Autonomous Legged and Aerial Robotic Exploration in the Tunnel and Urban Circuits of the DARPA Subterranean Challenge. Field Robotics, 2, 274–324, 2022.

**[J4] H. Nguyen** and K. Alexis. Forceful Aerial Manipulation Based on an Aerial Robotic Chain: Hybrid Modeling and Control. IEEE Robotics and Automation Letters, 6(2), 3711–3719, 2021.

### Conference and Proceedings Papers

**[C1]** M. Dharmadhikari, P. De Petris, M. Kulkarni, N. Khedekar, **H. Nguyen**, A. E. Stene, E. Sjøvold, K. Solheim, B. Gussiaas and K. Alexis. Autonomous Exploration and General Visual Inspection of Ship Ballast Water Tanks Using Aerial Robots. International Conference on Advanced Robotics (ICAR), 2023.

**[C2]** M. Kulkarni, **H. Nguyen** and K. Alexis. Semantically-Enhanced Deep Collision Prediction for Autonomous Navigation Using Aerial Robots. IEEE/RSJ IROS, 3056–3063, 2023.

**[C3]** M. R. Dharmadhikari, P. De Petris, **H. Nguyen**, M. Kulkarni, N. Khedekar and K. Alexis. Manhole Detection and Traversal for Exploration of Ballast Water Tanks Using Micro Aerial Vehicles. International Conference on Unmanned Aircraft Systems (ICUAS), 2023.

**[C4]** P. De Petris, S. Khattak, M. Dharmadhikari, G. Waibel, **H. Nguyen**, M. Montenegro, N. Khedekar, K. Alexis and M. Hutter. Marsupial Walking-and-Flying Robotic Deployment for Collaborative Exploration of Unknown Environments. IEEE SSRR, 188–194, 2022.

**[C5]** P. De Petris, M. Dharmadhikari, **H. Nguyen** and K. Alexis. Risk-Aware Motion Planning for Collision-Tolerant Aerial Robots Subject to Localization Uncertainty. IEEE/RSJ IROS, 4561–4568, 2022.

**[C6]** P. De Petris, **H. Nguyen**, M. Dharmadhikari, M. Kulkarni, N. Khedekar, F. Mascarich and K. Alexis. RMF-Owl: A Collision-Tolerant Flying Robot for Autonomous Subterranean Exploration. ICUAS, 536–543, 2022.

**[C7]** M. Kulkarni, M. Dharmadhikari, M. Tranzatto, S. Zimmermann, V. Reijgwart, P. De Petris, **H. Nguyen**, N. Khedekar, C. Papachristos, L. Ott, R. Siegwart, M. Hutter and K. Alexis. Autonomous Teamed Exploration of Subterranean Environments Using Legged and Aerial Robots. IEEE ICRA, 3306–3313, 2022.

## Publications continued

### Conference and Proceedings Papers

**[C8] H. Nguyen**, S. H. Fyhn, P. De Petris and K. Alexis. Motion Primitives-based Navigation Planning Using Deep Collision Prediction. IEEE ICRA, 9660–9667, 2022.

**[C9] H. Nguyen**, M. Kamel, K. Alexis and R. Siegwart. Model Predictive Control for Micro Aerial Vehicles: A Survey. European Control Conference (ECC), 1556–1563, 2021.

**[C10]** M. R. Dharmadhikari, **H. Nguyen**, F. Mascarich, N. Khedekar and K. Alexis. Autonomous Cave Exploration Using Aerial Robots. ICUAS, 942–949, 2021.

**[C11]** P. De Petris, **H. Nguyen**, M. Kulkarni, F. Mascarich and K. Alexis. Resilient Collision-Tolerant Navigation in Confined Environments. IEEE ICRA, 2286–2292, 2021.

**[C12]** F. Mascarich, P. De Petris, **H. Nguyen**, N. Khedekar and K. Alexis. Autonomous Distributed 3D Radiation Field Estimation for Nuclear Environment Characterization. IEEE ICRA, 2163–2169, 2021.

**[C13]** P. De Petris, **H. Nguyen**, T. Dang, F. Mascarich and K. Alexis. Collision-Tolerant Autonomous Navigation through Manhole-Sized Confined Environments. IEEE SSRR, 84–89, 2020.

**[C14]** S. Khattak, **H. Nguyen**, F. Mascarich, T. Dang and K. Alexis. Complementary Multi-Modal Sensor Fusion for Resilient Robot Pose Estimation in Subterranean Environments. ICUAS, 1024–1029, 2020.

**[C15]** M. Kulkarni, **H. Nguyen** and K. Alexis. The Reconfigurable Aerial Robotic Chain: Shape and Motion Planning. IFAC-PapersOnLine, 53(2), 9295–9302, 2020.

**[C16] H. Nguyen**, T. Dang and K. Alexis. The Reconfigurable Aerial Robotic Chain: Modeling and Control. IEEE ICRA, 5328–5334, 2020.

**[C17]** M. R. Dharmadhikari, T. Dang, L. Solanka, J. B. Loje, **H. Nguyen**, N. Khedekar and K. Alexis. Motion Primitives-based Path Planning for Fast and Agile Exploration Using Aerial Robots. IEEE ICRA, 179–185, 2020.

**[C18]** F. Mascarich, **H. Nguyen**, T. Dang, S. Khattak, C. Papachristos and K. Alexis. A Self-Deployed Multi-Channel Wireless Communications System for Subterranean Robots. IEEE Aerospace Conference, 1–8, 2020.

**[C19]** T. Dang, F. Mascarich, S. Khattak, **H. Nguyen**, Hai Nguyen, S. Hirsh, R. Reinhart, C. Papachristos and K. Alexis. Autonomous Search for Underground Mine Rescue Using Aerial Robots. IEEE Aerospace Conference, 2020.

**[C20]** T. Dang, F. Mascarich, S. Khattak, **H. Nguyen**, N. Khedekar, C. Papachristos and K. Alexis. Field-hardened Robotic Autonomy for Subterranean Exploration. Conference on Field and Service Robotics (FSR), 2019.

### Preprint

**[P1]** Khang H. Nguyen, Hoang Pham Quang Nguyen, Ha Phuong Nguyen, Khanh Dinh Binh, Xuan Ha Nguyen, Vien Ngo, Duy Ho Nguyen Minh, **Huan Nguyen** and An T. Le. FINE: Future-Informed Navigation Encoding for Data-Efficient Vision-Language Navigation. arXiv:2609.32855, 2026. Preprint.